BEING SYNTOMIC IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Characterisation of syntomic morphisms by local finite presentation, flatness and local complete intersection fibres. Morphisms, the syntomic chapter's characterisation, quoted at proof lines 6376-6377. Splits syntomicity into three properties.
- Flatness and local finite presentation are fpqc local on the base. Same chapter, Lemmas lemma-descending-property-flat and lemma-descending-property-locally-finite-presentation, invoked at proof lines 6378-6381. Gives two of the three ingredients.
- The locus where the fibres are local complete intersections is fpqc local on the base. Morphisms, Lemma morphisms-lemma-set-points-where-fibres-lci, invoked at proof lines 6382-6383. Gives the third ingredient and hence the result.

Source reference. Original lemma 02L2: sources/stacks-descent/pinned-original-source.tex, source lines 6050–6050. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KY: sources/stacks-descent/pinned-original-source.tex, source lines 5984–5984. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02V3: sources/stacks-chow/tags.txt, source lines 3754–3754. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is syntomic" is fpqc local on the base.*

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Proof. A morphism is syntomic if and only if it is locally of finite presentation, flat, and has locally complete intersections as fibres. We have seen already that being flat and locally of finite presentation are fpqc local on the base (Lemmas 02L2, and 02KY). Hence the result follows for syntomic from Morphisms, Lemma 02V3. □